

PRACTICAL 3

Aim : Prepare and Test Straight UTP Cable and Cross UTP Cable.

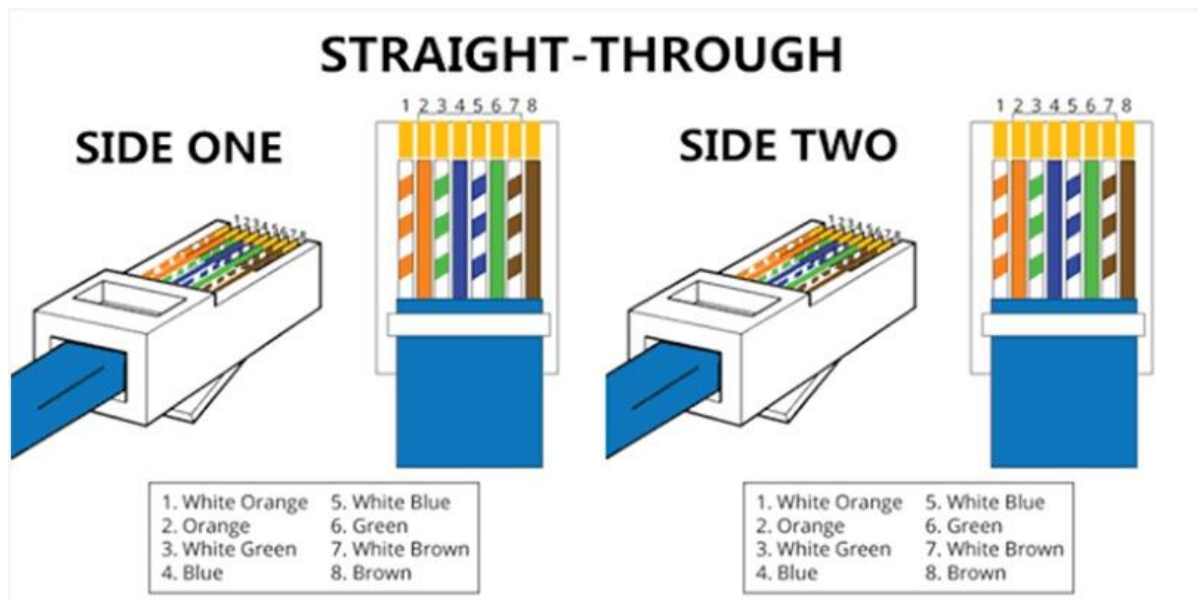
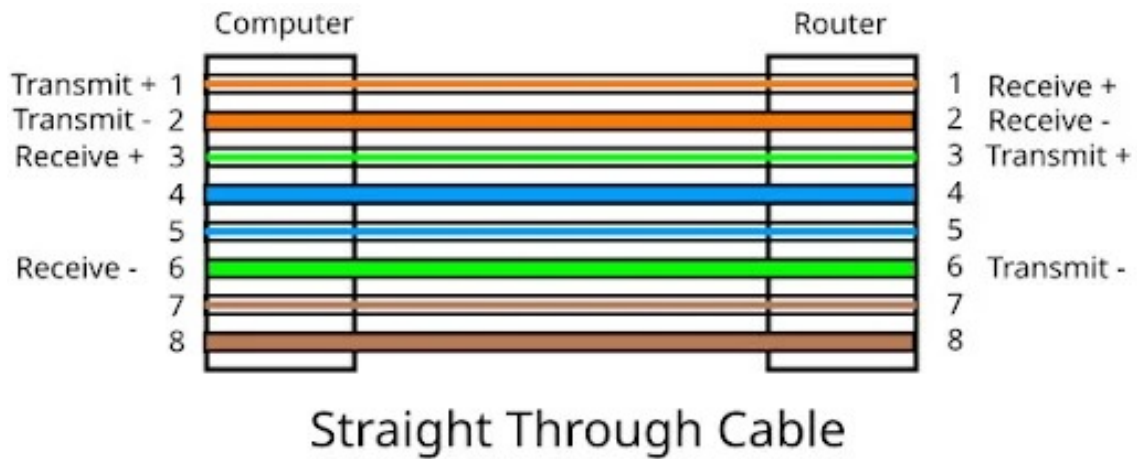
Input :

What is Straight Through Cable?

- A straight-through cable is a type of Ethernet cable used to connect different types of devices within a network.
- When connecting between two dissimilar devices, for example, from the computer to the router, a straight-through cable will be needed. There are no left or right constrictions on the straight-through cable, as both ends have the same wiring signals, so the wires correspond to the same pins at the other end, too. In such a way, the transceiver protocol can be set up in such a manner that the information is transmitted from one device to another without any need for manual configuration.

Features of Straight-Through Cable -

- **Direct Connection :** Links devices of different categories for instance a computer with a switch or a router, Computer-to-Device connection.
- **Standard Ethernet Cable :** It is particularly used in Ethernet networks for data transmission purposes.
- **Color Code Consistency :** The endings of the electrodes have identical sequences in terms of the color codes applied to the wires.
- **Common Usage :** Sometimes used in home and office networks for sharing devices among the nodes and link points.

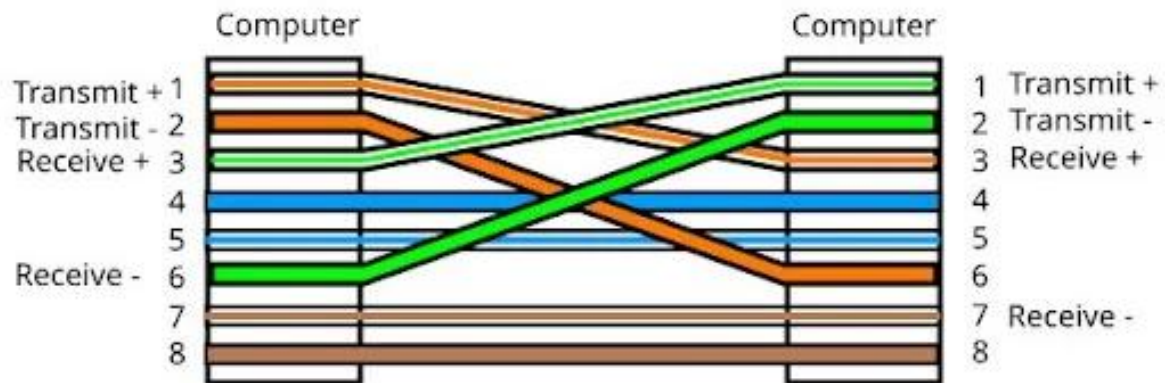


What is a Crossover Cable?

- A crossover cable is a type of Ethernet cable used to connect similar devices directly to each other without needing an intermediate network device like a switch or hub.
- Crossover cables are found primarily on connections between similar devices like two computers or two switches. A difference between straight-through and crossover cables is that the order of the wires differs at each end of the cable. This will allow communication of devices indicating if they are transmitting or receiving by 'crossover'ing the 'Tx' and 'Rx' so that data can be obtained by both similar devices.

Features of Crossover Cable -

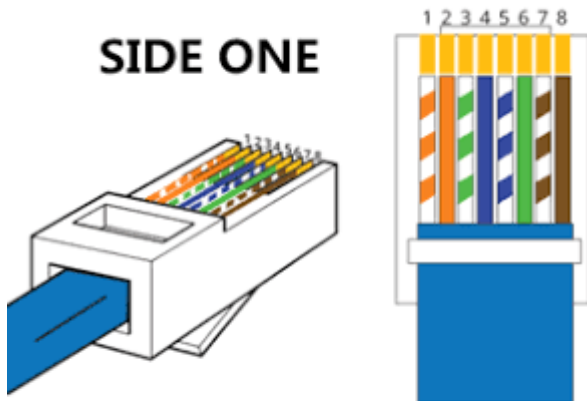
- **Device-to-Device Connection** : Mostly used in direct connectivity between composites of similar types like two computers or two switches.
- **Pin Configuration** : Pins connected on two ends with different arrangements for example T568 A on one end and T568B on the other end.
- **Transmit/Receive Swapping** : Exchange the two ports' T/R signal between the two ends.
- **Specific Purpose** : Typically employed in scenarios where devices must be able to interface without requiring a hub or switch.
- **Colour Code Variation** : In a crossover, some wires at each end have colour-coded sequences different from the other end.



Crossover Cable

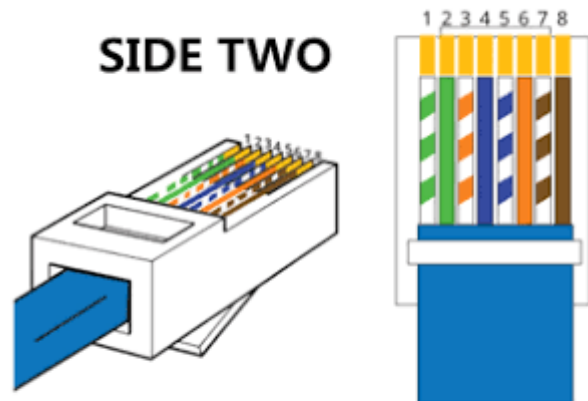
CROSSOVER

SIDE ONE



- | | |
|-----------------|----------------|
| 1. White Orange | 5. White Blue |
| 2. Orange | 6. Green |
| 3. White Green | 7. White Brown |
| 4. Blue | 8. Brown |

SIDE TWO



- | | |
|-----------------|----------------|
| 1. White Green | 5. White Blue |
| 2. Green | 6. Orange |
| 3. White Orange | 7. White Brown |
| 4. Blue | 8. Brown |